

The Effects of Visual and Verbal Priming on the Interpretation of Ambiguous Images

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Introduction

“Priming is a nonconscious form of human memory, which is concerned with perceptual identification of words and objects.” (Tulving & Schacter, 1990, p. 301) Priming, which is also commonly defined as the influence of prior exposure to certain stimuli, allows subtle cues to influence decision-making. (Shapir et al., 2023) Nevertheless, compared with its influence on decision-making, the extent to which priming affects early perceptual interpretation remains less understood. Considering the potential application of priming in fields like advertising, human-computer interaction, and psychological assessments, the current study aims to investigate whether priming influences our interpretation of visual stimuli, and whether the format of the stimuli matters.

Shapir et al. 's (2023) study serves as an example of how priming influences decision-making in a real-world setting. In their study, they found that students exposed to pizza-related cues in a lecture were more likely to choose pizza for lunch. The experimental design included four conditions: no prime, visual primes (pictures in slide backgrounds), auditory primes (pizza mentioned as a teaching example), and a combined audiovisual condition. All priming groups chose pizza more often than the control group and the combined condition had the strongest effect. Notably, the effect varied by gender and culture—females responded more to visual primes, and Chinese students differed from American students in their responses. These findings reveal that priming works outside the lab, and that its strength depends on how the prime is delivered and who receives. However, since the outcome measure was a deliberate choice between food options, this study speaks primarily to priming's influence on decision-making rather than on perceptual interpretation itself.

To test whether priming can affect perception, we examined the memory color effect in Cohen et al. (2026). This effect asks whether prior knowledge can influence perception of

ambiguous input. In their experiment, participants viewed completely desaturated grayscale images of familiar objects—stop signs, bananas, and the American flag. These images were shown under two lighting conditions: extremely dim, ambiguous light and bright, unambiguous light. Under dim light, most participants reported faint canonical colors. For example, a gray stop sign looked reddish. This effect nearly vanished under bright light. One major concern, however, was demand characteristics—participants complying with the experimenter. To rule this out, Cohen et al. ran a control study. Participants viewed two images side by side: one grayscale, one color-rotated. Under dim light, neither matched the object's true color. Nonetheless, participants consistently rated the grayscale image as more colorful. Critically, a separate group who only heard about the setup showed no such preference. This suggests that effect was perceptual, not driven by guessing or compliance, and provides support for the idea that prior knowledge can influence the initial interpretation of ambiguous visual stimuli.

While Cohen et al. demonstrated that long-term knowledge can shape perception, whether unconscious semantic information can achieve the same effect requires a closer look into semantic priming. Bruno et al. (2020) examined whether subliminal semantic primes, in either picture or word format, could speed up a categorization task. They used a masked priming paradigm (masking pictures with pictures and words with words), presenting the stimuli for only an instant to ensure unconsciousness. Participants categorized targets as animals or objects while prime-target similarity varied. To rule out potential confounds in masking procedures, they also ran a control experiment in which verbal primes were masked with pictures. The pattern of results did not differ much from the main experiment: pictures were categorized faster than words, but the strength of the priming effect was similar across the two formats. Additionally, this study had limitations in its generalizability since only female participants were studied. To avoid the same trap, our study will include samples of

both male and female while implementing similar methodologies. Having established that unconscious semantic priming can influence behavior, we next asked whether it can also influence perception—specifically, the interpretation of ambiguous visual input in the form of verbal and pictorial primes mentioned above.

Moreover, Bussche et al. (2009) conducted a meta-analysis of 54 masked priming studies to identify the boundary conditions of unconscious semantic priming. They found that subliminal primes are indeed processed semantically, even when primes are novel and drawn from large categories. However, the effect is stronger when stimulus-response mappings are available, and this leads us to question whether the result is influenced by the stimulus-response mappings. To eliminate confounding variables, three methodological considerations are summarized out of the key moderators mentioned by Bussche et al. (sample size, target set size, prime duration, correspondence between target and primes, and whether or not prime visibility was measured). First, prime visibility must be measured objectively -- using a sensitivity measure like d' , rather than relying on subjective reports -- to confirm that any observed effects arise from unconscious processing. Second, primes and targets must be semantically related for a more observable, robust activation. Third, using novel primes that never appear as targets helps prevent the formation of automatic stimulus-response mappings that could produce priming without semantic processing. This meta-analysis provided clear methodological guidance on testing our research topic and further narrowed our scope to the field of unconscious semantic priming.

The studies reviewed above establish two important points: prior knowledge can influence perception under ambiguity, and unconscious semantic priming can be robust and measurable. Several questions still remain unexplored though. For example, will individuals previously exposed to bird-related primes be more likely to identify the duck-rabbit image as a duck at first view than those exposed to mammal-related primes? Will the format of the

primes (e.g. pictures and words) influence this effect? Our goal is to study whether semantic primes can alter the initial interpretation of an ambiguous visual stimulus and whether the effect differs between word-based and image-based cues. Word primes produced smaller effects than symbol primes when format was tested by itself Bussche et al. (2009) stated. On account of that, we hypothesize that unconscious semantic priming can alter the initial interpretation of an ambiguous visual stimulus, and pictorial priming will have a stronger influence than verbal priming. This hypothesis will be tested by an experiment with participants randomly distributed to different priming conditions.

Method

Participants

The participants consisted of 51 individuals whom researchers reached out to personally. Participant ages ranged from 16 to 49 years old, with the median age being 17. Said participants were of both genders, with 33 participants from China, 1 from Korea, 1 from Brazil, and 15 from the US. Participants were not selected based on nationality or age.

Materials

We used a Windows Laptop to embed a series of photos to prime participants through Psytoolkit (Stoet, 2010, 2017). The image mask was generated by python using a random pixel noise algorithm implemented through NumPy library. The prime image for the Duck condition was from the THINGS database of 1,854 object concepts and more than 26,000 naturalistic object images (Herbart et al., 2019). The ambiguous visual stimulus depicted is the duck-rabbit illusion, chosen from figure 20 of *The Mind's Eye* (Jastrow, 1899). All images were cropped to 600*600 in resolution to standardise into the survey.

Procedure

We recruited participants by reaching out to them individually and sending them a link to the online survey. The survey and experiments were implemented and presented online using the PsyToolkit platform (Stoet, 2010, 2017). On the first page, participants read a short welcome message stating that all data collected would be completely anonymous and used for a short research paper, that the survey would take no longer than five minutes, and that once started, the survey should not be exited until the entire form was completed. On the next three pages, participants reported their nationality, gender, and age, respectively.

Participants then read the instructions for the task: “You will now see a brief flash followed by an image. Please answer based on your first impression.” When ready, participants clicked a button to begin. In the word prime condition, participants saw a former mask (&&&&&) for 100 ms, followed by the prime word “DUCK” for 33 ms, the same prime duration used by Bruno et al. (2020), followed by a latter mask (#####) for 33 ms. In the image prime condition, participants saw a pixel-noise mask for 100 ms, followed by an image of a duck for 1 ms, followed by a second pixel-noise mask for 33 ms. The 1 ms duration was chosen so that the duck image would reach the eye without allowing enough time for it to be consciously processed. As an offset for the disparity between prime durations, the word prime is displayed with a black background identical to that of its symbolic masks, mitigating the participant’s ability to distinguish between them. In the control condition, participants saw a blank black screen for 166 ms, matching the total duration of the word prime sequence so that the illusion appeared after a comparable delay in all three groups. PsyToolkit’s counterbalancing function assigned participants to one of the three conditions, ensuring a random and even split between the groups.

Immediately after the prime or black screen, the duck-rabbit illusion appeared on screen and remained visible until the participants responded by typing out what they saw first. On the following page, participants were asked whether they could identify anything

meaningful in their respective priming condition. This step was essential to ensuring that the prime was perceived unconsciously. The illusion was then shown a second time, and participants were asked whether they were able to identify any other animals in the image. Each participant completed the survey only once; no practice trials were given due to the brevity of the experiment and possibility of affected answers from repeated exposure.

Results

Initial Interpretation

Figure 1

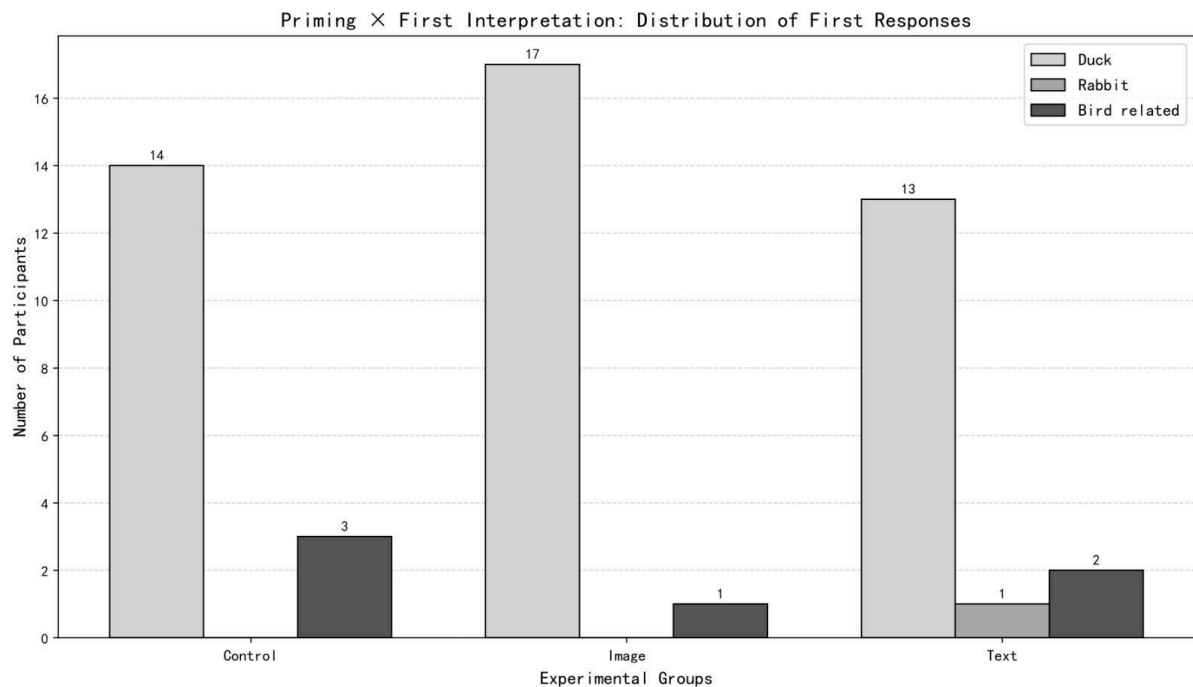
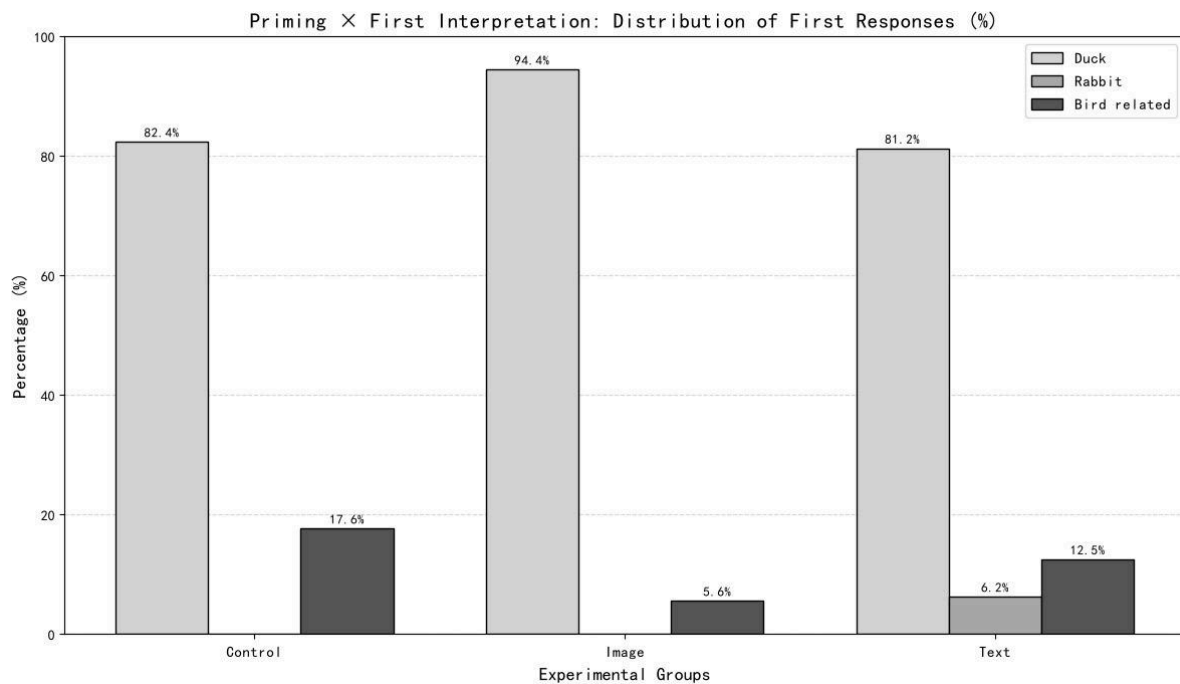


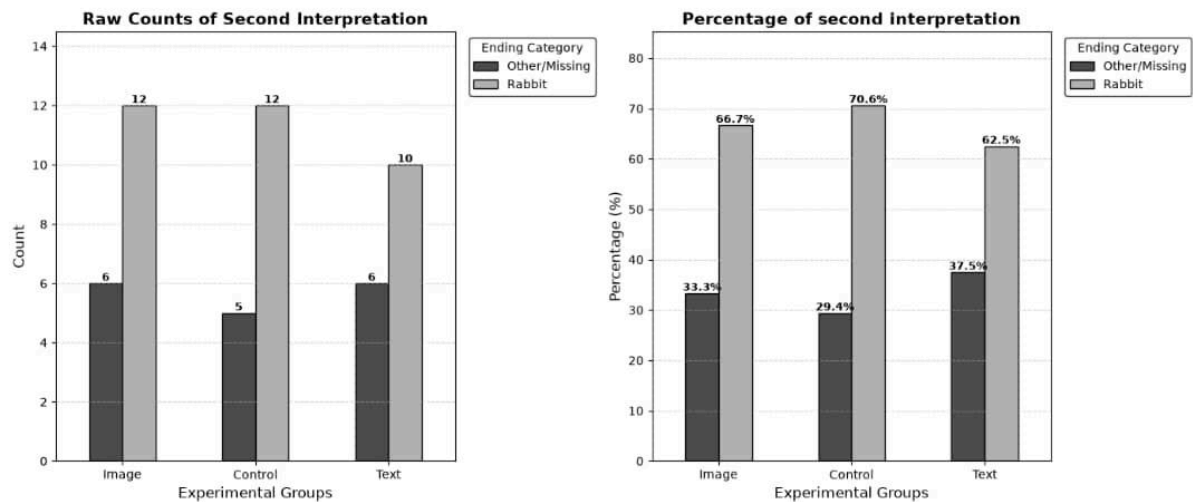
Figure 2



Our results for the first interpretation showed that Duck was the most common response in every priming condition. Our main hypothesis was that the Image prime would influence first interpretations more strongly than the Text prime. As shown in Figure 1 and 2, 14 participants (82.4%) in the Control condition ($n = 17$) reported Duck first, along with 17 (94.4%) in the Image condition ($n = 18$) and 13 (81.2%) in the Text condition ($n = 16$). Rabbit was almost never the first response. No participant in the Control or Image condition reported it first (0.0% in each), and 1 participant (6.2%) in the Text condition did. The remaining responses were Bird-related (eg. “crow”): 3 (17.6%) in Control, 1 (5.6%) in Image, and 2 (12.5%) in Text. With a vast majority of ducks initially interpreted, our hypothesis is partially supported, though it is unclear whether word-priming has an effect due to its relatively low number of duck interpretations.

Other Interpretations

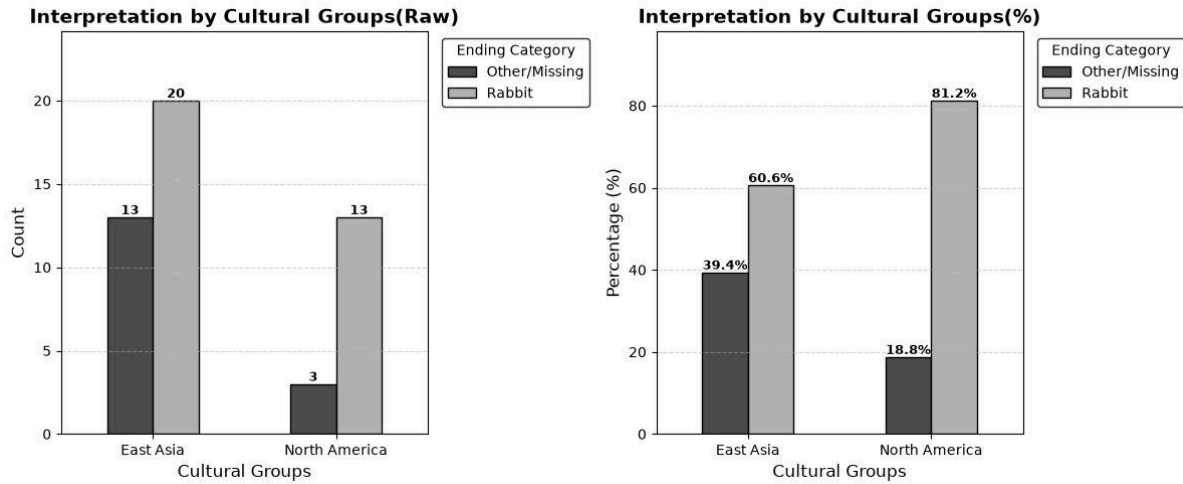
Figure 3



To understand whether priming inhibited participants' ability to identify the rabbit that is unprimed for, we asked for second interpretations. We predicted that the two primed groups would be less likely to do so than the Control group. As shown in Figure 2, 12 participants (70.6%) in the Control condition ($n = 17$) also reported the Rabbit, along with 12 (66.7%) in the Image condition ($n = 18$) and 10 (62.5%) in the Text condition ($n = 16$). The remaining responses fell into the Other/Missing category: 5 (29.4%) in Control, 6 (33.3%) in Image, and 6 (37.5%) in Text. It is important to note that the Other/Missing identification category consisted of two non-bird identifications, one duck identification, various identifications of bird-related species, and many that were not able to identify other animals. Both primed groups fell below Control in Rabbit reporting, which matched the direction we predicted. We therefore counted this pattern as modest support for our hypothesis.

Cultural Differences

Figure 4



Note: To maintain balance and due to small sample size, one Brazilian participant was excluded for this visualization.

For the cultural comparison, we measured how often participants in each cultural group reported the Rabbit. As shown in Figure 3, 20 participants (60.6%) in the East Asia group ($n = 33$) reported the Rabbit, and 13 (39.4%) fell into the Other/Missing category. In the North America group ($n = 16$), 13 participants (81.2%) reported the Rabbit, and 3 (18.8%) fell into Other/Missing. The North America group reported the Rabbit at a clearly higher rate. We concluded that this disparity could be a result of geographic location and a relatively higher rabbit population in the Americas.

Discussion

The minor effect observed in the image-prime condition is broadly consistent with Shapir et al. (2023), who showed that priming can influence cognition outside tightly controlled lab settings. However, we found no observable effect for word primes, which contrasts with the verbal priming effects reported by Bussche et al. (2009) and Bruno et al. (2020). Visual primes may be processed more directly than verbal cues. The higher level of abstraction of words takes humans longer to process than images. (Bruno et al., 2020), making them more

likely to influence perceptual judgment. This finding about initial interpretations is consistent with Cohen et al.'s (2026) conclusion, which suggests that prior knowledge can shape perception when sensory input is ambiguous. However, unlike Cohen et al.'s study, where ambiguity resulted from degraded sensory information, the present study involved a stimulus containing two competing interpretations, which can account for the relatively limited priming effect.

Priming also appeared to inhibit access to the alternative interpretation. Participants exposed to duck-related primes were less likely to report the rabbit interpretation when asked to identify another animal. Similar to Cohen et al.'s (2026) finding that prior knowledge can influence ambiguous sensory information, activating a duck-related concept may have increased the accessibility of the duck interpretation while reducing the likelihood that participants would switch to the rabbit interpretation.

The cultural comparison revealed that North American participants reported the rabbit interpretation more frequently than East Asian participants, implying a stronger priming effect for ducks on East Asian participants. This could be explained by differences in familiarity with these animals, which may influence interpretation. For example, rabbits may have greater cultural visibility in some North American contexts through Easter-related imagery, while ducks may have greater familiarity in some East Asian contexts through everyday exposure. These differences in long-term exposure are likely to influence which interpretation becomes more accessible when viewing ambiguous stimuli. Similarly, Shapir et al. (2023) also found that priming effects can vary across cultural groups. However, these explanations remain speculative because familiarity was not directly measured.

Limitations arose from the online platform. The image and word primes were presented for different durations. This difference was caused by technical limitations of the online

platform, as longer image presentation made the prime consciously visible. Therefore, the difference between image and word priming effects may partly reflect presentation duration rather than stimulus type alone. The online experimental setting also reduced control over factors such as screen performance, viewing distance, and participant attention, which have introduced confounds.

Furthermore, there are limitations concerning the ambiguous stimulus itself. Although the duck–rabbit illusion is widely used to study perceptual ambiguity, the image may not provide completely equal perceptual accessibility for both interpretations. The observed differences between conditions are related to inherent characteristics of the stimulus rather than the influence of priming alone.

Moreover, due to time constraints, the study relied on a single self-report consciousness check without additional measures for triangulation, such as objective awareness tests or quantitative sensitivity measures. Therefore, although most participants appeared unaware of the primes, unconscious processing cannot be confirmed with complete certainty. Furthermore, even some participants in the blank control condition reported seeing meaningful information., This reveals that some participants have interpreted the awareness question in an unexpected way or had difficulty distinguishing meaningful stimuli from other visual experiences.

Bringing the paradigm into a controlled lab setting, with matched presentation durations and more quantitative measures like reaction time or eye tracking, would help clarify these effects. Additionally, directly measuring participants' familiarity with culturally relevant objects will help clarify whether environmental exposure contributes to differences in ambiguous perception. Examining other ambiguous figures would also help determine whether the influence of unconscious priming generalizes beyond the duck–rabbit illusion.

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